

Leart Zuka

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🌐 LinkedIn

🐙 Github

🌐 Website

Summary

Physics master's student based in Munich, Germany, with hands-on experience in quantum optics research, FPGA-based lab automation, and full-stack software development, including a thesis on atom-photon quantum gates at the Max Planck Institute for Quantum Optics, and a track record spanning peer-reviewed publication, student leadership, and industry software roles.

Employment History

- Apr. 2022 – Nov. 2023 📌 **Software Developer**, Titanom Technologies GmbH., Munich
Designed and implemented back-end APIs and databases (Python, PostgreSQL) and full-stack web applications (JavaScript/TypeScript, React, Java).
- Apr. 2024 – Apr. 2025 📌 **Research Assistant**, Max Planck Institute for Quantum Optics, Munich
Programmed FPGA-based laser control sequences for trapping and cooling of ^{87}Rb atoms, and developed real-time monitoring/analysis software contributing to a peer-reviewed publication.

Education

- 2024 – 2026 📌 **M.Sc. Condensed Matter Physics** at Technical University Munich
Thesis title: *An atom-photon quantum gate using an atomic clock qubit and a birefringent cavity.*
Grade: Pending (defense completed July 2026)
- 2020 – 2023 📌 **B.Sc. Physics** at Technical University Munich.
Thesis title: *Analysis of X-ray/Neutron Scattering Data Using Bornagain.*
Grade: 2.5

Research Publications

Journal Articles

- 1 G. Chiarella, T. Frank, L. Zuka, P. Farrera, and G. Rempe, "Source of heralded atom-photon entanglement for quantum networking," *Phys. Rev. Lett.*, vol. 135, p. 240 802, 24 Dec. 2025. 🌐 DOI: 10.1103/5zk9-3rpv

Skills

- Languages 📌 Native German Speaker; English (C2); French (B1).
- Technical Skills 📌 Linux, Python, Bash, FPGA (Keysight), Qiskit, Javascript, Typescript, C, CSS, Git, $\text{\LaTeX}$
- Soft Skills 📌 Team Leadership, Project Management, Technical Communication, Cross-functional Collaboration, Mentoring, Problem Solving.

Miscellaneous Experience

Extra-curricular activities

- Jul. 2024 – Jun. 2026 📌 **IT Department Head**, PushQuantum e.V.; Project Manager, PushQuantum WiQi

Awards and Achievements

- Nov. 2024 📌 **IBM Bootcamp Winner**. 2024 Qiskit Fall Fest organized by IBM x PushQuantum e.V.